Question 5		
(a)	Electric field between plates M and N $E = \frac{V}{d} = \frac{235-(-235)}{2.00\times10^{-2}} = 2.35\times10^{4}\ \text{V m}^{-1}$ $F = Eq = \left(2.35\times10^{4}\right)\left(58\times10^{-12}\right) = 1.4\times10^{-6}\ \text{N}$	**B1** **A1**
(b)	Acceleration, $a = \frac{F}{m} = \frac{1.363\times10^{-6}}{0.12\times10^{-3}} = 1.136\times10^{-2} = 1.1\times10^{-2}\ \text{m s}^{-2}$	**A1**
(c)	Time to travel from M to N $s = ut + \frac{1}{2}at^{2}$ $2.0\times10^{-2} = 0 + \frac{1}{2}\left(1.136\times10^{-2}\right)\left(t^{2}\right)$ $t = 1.8765\ s$ Frequency, $f = \frac{1}{2\times t} = \frac{1}{2\times1.8765} = 0.266\ \text{Hz}$	**B1** **B1** **A1**
(d)	The acceleration for simple harmonic motion is proportional to its displacement. Since the acceleration of the sphere is constant, the motion cannot be simple harmonic.	**A1**